The Algorithm and Main Result

$\min: f(x)$

Phase I and II: damped and undamped Newton

Phase I: $x_{t+1} = x_t - \eta \left(\nabla^2 f(x_t)\right)^{-1} \nabla f(x_t)$

Phase II: $x_{t+1} = x_t - \left(\nabla^2 f(x_t)\right)^{-1} \nabla f(x_t)$

quadratic convergence

Picking step size: back tracking line search

Fix $0 < \alpha < 1/2$, $\beta < 1$

BTLS:

$$(*) \quad f(x_t + \eta (\nabla^2 f(x_t))^{-1} \nabla f(x_t)) \leq f(x_t) - \alpha \cdot \eta \left(\nabla f(x_t)^T \nabla^2 f(x_t)^{-1} \nabla f(x_t) \right)$$

start with $\eta = 1$

if not: $\eta \leftarrow \eta\beta$

The Main Theorem

Suppose f convex, m- strongly convex, M-smooth

also: $\nabla^2 f$ L - Lipschitz (new!)

$$\|\nabla^2 f(x) - \nabla^2 f(y)\|_{op} \leq L\|x-y\|_2$$

(Def: $\|M\|_{op} = \sigma_{max}(M)$, if M symmetric $= |\lambda_{max}(M)|$)

Then: $\exists$ constants λ, $0 < \lambda \leq \frac{m}{2L}$, $\gamma = \alpha\beta \cdot \lambda^2 \cdot \frac{m}{M^2}$

$f(x_t) - f(x_{t+1}) \geq \gamma$

s.t.: If $\|\nabla f(x)\| \geq \lambda$ (i.e., if we are not close to x^*)

we are in Phase I and use damped Newton with BTLS

If $\|\nabla f(x)\| < \lambda$, we are in Phase II and can use step size $= 1$

and $$\|x_{t+1} - x^*\|_2 \leq \frac{L}{m}\|x_t - x^*\|^2$$

Discussion: Phase I and Phase II

Phase I : $f(x_t) - f(x_{t+1}) > \gamma > 0$

$\Rightarrow$ we are in Phase I for a finite # steps.

Phase II : we enter when $\|\nabla f(x)\| \leq \lambda$

strong convexity $\Rightarrow$ quad lower bound. Therefore

$\|\nabla f(x)\|$ small $\Rightarrow$ $\|x - x^*\|$ small.

$$f(x^*) \geq \underbrace{f(x) + \nabla f(x)^T (x^* - x)} + \underbrace{\frac{m}{2} \|x^* - x\|_2^2}$$

$$\Rightarrow \quad \frac{m}{2} \|x^* - x\|_2^{\cancel{2}} \leq \underbrace{\left(f(x^*) - f(x) \right)} + \nabla f(x)^T (x - x^*)$$

$$\leq \|\nabla f(x)\| \cdot \cancel{\|x - x^*\|}$$

$$\|x^* - x\|_2 \leq \frac{2}{m} \|\nabla f(x)\|_2$$

Phase II: quadratic convergence

Thm promises: if $\|x_t - x^*\| \leq \frac{m}{2L}$

for $x_{t+1} = x_t - (\nabla^2 f(x_t))^{-1} \nabla f(x_t)$

$$\|x_{t+1} - x^*\| \leq \frac{L}{m} \|x_t - x^*\|^2 \leq \frac{L}{m}\left(\frac{m}{2L}\right)^2 = \overbrace{\frac{1}{2}\left(\frac{m}{2L}\right)}$$

$$\|x_{t+2} - x^*\| \leq \frac{L}{m} \|x_{t+1} - x^*\|^2 \leq \left(\frac{1}{2}\right)^3 \cdot \frac{m}{2L}$$

$$\|x_{t+3} - x^*\| \leq \frac{L}{m} \|x_{t+2} - x^*\|^2 \leq \left(\frac{1}{2}\right)^7 \cdot \frac{m}{2L}$$

$$\|x_{t+4} - x^*\| \leq \quad \leq \left(\frac{1}{2}\right)^{15} \cdot \frac{m}{2L}$$

$$\vdots$$

$$\|x_{t+k} - x^*\| \leq \quad \leq \underbrace{\left(\frac{1}{2}\right)^{2^k - 1}} \cdot \left(\frac{m}{2L}\right)$$

<u>Linear</u> : $\left(\frac{1}{2}\right)^t$, <u>Quadratic</u>: $\left(\frac{1}{2}\right)^{2^t}$